



Evaluate profiles

Project a chain onto a plane and cut it where it folds back -- or outline a part and split its outline into top, bottom and middle profiles.



What it does

Side profiles of a core, a topsheet or a sidewall are needed as clean curves: for fills, for drawings, for measuring. Native Project and Outline give an undivided loop, with slivers, that can fold back at the tip. Evaluate profiles gives the profiles you actually want, named.

Use it when

- You need a part's top / bottom / middle side profile
- A projected chain folds back at a tip
- An outline has micro-edges that break a fill

P1: a crowned sheet (3 mm high across its width) outlined onto Front: top, bottom and middle profiles (height x60)



P1: a crowned sheet outlined: top, middle and bottom profiles.

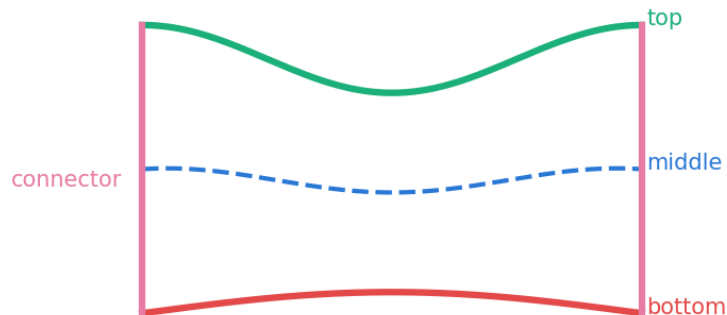


Two jobs

Edges: the projected chain is cut where it doubles back



Part: the outline split along the prevailing direction



Edges

Project, then cut where the chain stops advancing along the prevailing direction (400-sample scan, then bisection).

Part

The two longest runs along the direction are top and bottom (top = higher); the middle is their average at matching stations.

Micro-edges merged

Outline edges under 0.01 mm are merged away -- a 1.7 um edge once broke a fill.



The dialog: Part (1/2)

P1 crowned sheet picked... ✓ ✗

1 Profile from **Part** ▼

2 Name **crown**

3 **Part or surface to outline**
Unknown of Crowned sheet, 1... ✕

4 Return **All profiles** ▼

5 Merge edges shorter than **0.01 ...**

6 Project onto
Front plane ✕

7 Prevailing direction  

8 Output **One curve** ▼

9 > Approximation

10 > Debug

 **Final**

#	Parameter	What it means
1	Profile from	EDGES: project a chain. PART: outline a solid or surface.
2	Name	Part outputs are named <name> top, <name> bottom, ...
3	Part or surface to outline	A solid or surface body, or a face of one.
4	Return	Top / Bottom / Middle / Periphery / All profiles / Full (profiles + connectors).
5	Merge edges shorter than	0.01 mm; 0 = off.



The dialog: Part (2/2)

P1 crowned sheet picked... ✓ ✗

1 Profile from **Part** ▼

2 Name **crown**

3 **Part or surface to outline**
Unknown of Crowned sheet, 1... x

4 Return **All profiles** ▼

5 Merge edges shorter than **0.01 ...**

6 **Project onto**
Front plane x

7 **Prevailing direction**  

8 Output **One curve** ▼

9 > Approximation

10 > Debug

 **Final**

#	Parameter	What it means
6	Project onto	A planar face.
7	Prevailing direction	The direction profiles run along (+ flip). Default: the longest in-plane extent.
8	Output	One curve / One per input curve (exact) / Efficient (merges exact lines and arcs; freeform only across tangent joints).
9	Approximation	60 samples, degree 3, 0.01 mm, 24 control points.
10	Debug	Scan, curve detail, colour the profiles.



The dialog: Edges

P10 chain, One per input ...  

1 Profile from **Edges** ▼

Name **chain per curve**

2 **Edges to project**
Chain on Top: lines 0>200>40... ×

3 Project onto
Top plane ×

4 Prevailing direction  

5 Output **One per input curve** ▼

> Approximation

> Debug

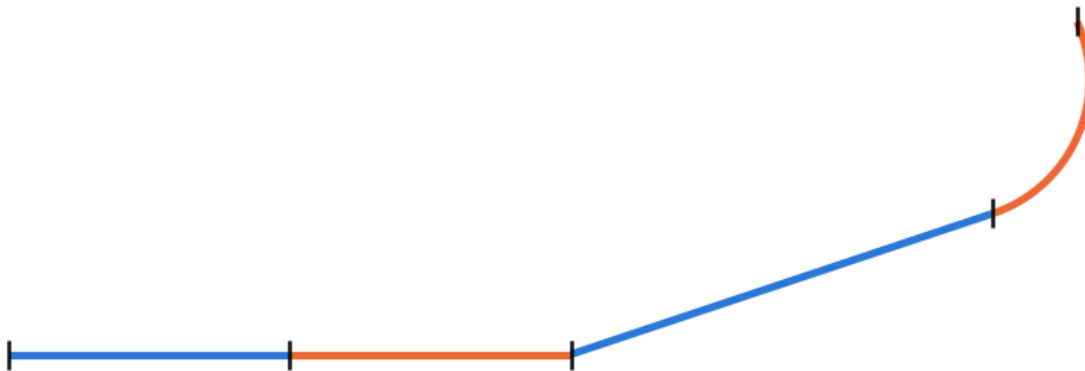
 **Final**

#	Parameter	What it means
1	Profile from	EDGES.
2	Edges to project	The chain.
3	Project onto	A planar face.
4	Prevailing direction	Default: the chain's chord.
5	Output	One curve / One per input curve / Efficient -- see the examples.



Example: One per input curve

P10 One per input curve: 4 exact edges (line, line, line, arc R100 cut where it turns back)



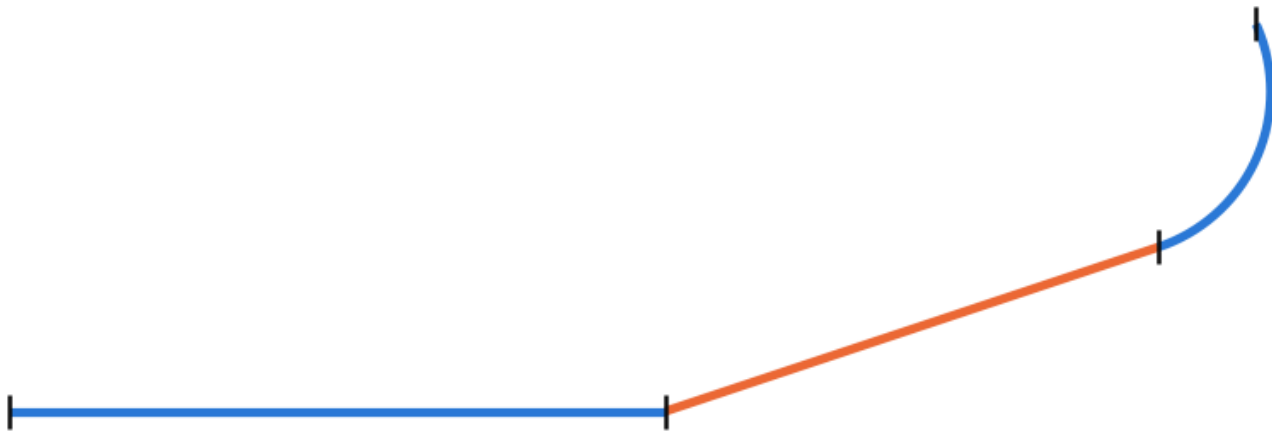
P10: lines 0-200-400 (collinear), a line to (700, 700), an R100 hook -- projected onto Top

4 exact edges: the projected edges themselves, the arc cut exactly where it turns back (at (759.55, 835.96)).



Example: Efficient

P11 Efficient: 3 edges -- the two collinear lines merged into one



P11: the same chain, Efficient

3 edges: the two collinear lines merged into one 400 mm line; the slanted line and the R100 arc kept exact. One curve (P9) gives a single fitted spline instead.



Outputs and messages

Published for Extract variables

Key	What it holds
top / bottom / middle	The named profile wires (empty when not requested).
periphery	The undivided outline (Return = Periphery).
connectors	The pieces joining top and bottom (Return = Full).
length	Length of the retained projection (the periphery for a part).
trimmedStart / trimmedEnd	Whether the projection was cut at each end.

Messages

	Message / what to do
Error	Nothing projected / not a single connected chain
Error	...seen edge-on: project its edges instead A surface lying in the projection direction has no outline.
Error	does not separate into a top and a bottom profile Check the prevailing direction.
Info	Merged N outline edge(s) shorter than ... Micro-edges removed; locations listed.



Tips and limits



Prefer exact output

One per input curve or Efficient keep lines and arcs exact; One curve is a fit.



Longest loop only

An outline with several loops keeps only the longest.



Efficient keeps corners

Freeform edges merge only across tangent joints (0.57 deg): a 28 deg corner stays a joint (P12), a tangent joint merges (P13).



Where it lives

Document	Curve_tools (2143812a99089658c704f0bc)
Tab	evaluate_profiles (uses curve_core)
Tests	Evaluate profiles tests studio: P1-P11 (P3 must error)
Examples	Evaluate profiles tests studio
Explainer	docs/explainers/curve_tools/curve_tools_explained.md, section 2.4
Related	Clean wire, Map curve, Merge curve, Evaluate profiles, Fillet wire; Extract variables